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CASE STUDY

Language Models for Value-Informed Proxy Decision Support

Victoria J. Nolan, Ph.D.,[1] Marcia MacGregor Brown, M.D., Ph.D.,[2] Jeremy A. Balch, M.D., Ph.D.,[1] Ankita V. Katukota,[3] Shayan Abbas,[1] Priyanka L. Devaguptapu, M.D.,[3] Naveen Baskaran, M.D., M.S.H.I.,[3] Kenneth L. Abbott, M.D.,[1] Philip Hong, M.D.,[1] Benjamin Shickel, Ph.D.,[4] Philip A. Efron, M.D.,[1] Gilbert R. Upchurch Jr, M.D., D.H.A.,[1] Azra Bihorac, M.D., M.S.,[4] Nathaniel Sharadin, Ph.D.,[5] Christopher J. Tignanelli, M.D., M.S., M.B.A.,[6,7] Ray E. Moseley, Ph.D.,[8] and Tyler J. Loftus, M.D., Ph.D.[1]

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Abstract

When patients lose decision-making capacity, large language models (LLMs) may help proxies choose treatments that honor incapacitated patients' values and preferences, especially when advance directives or formally identified proxies are ambiguous or unavailable. In a primary analysis, we prospectively enrolled 30 adults representing 15 dyads (15 patients alongside their legal, patient-designated proxy decision-maker) and compared patient–proxy and patient–LLM binary decisional agreement on multiple-choice hypothetical clinical scenarios, with and without providing patients' value profiles to the LLM; patients' value profiles were not provided to the human proxy. We found that patient–proxy decisional agreement (i.e., choosing the same treatment) on hypothetical clinical scenarios was 44.7% (95% confidence interval [CI], 28.6 to 61.9), similar to previous reports. Patient–LLM agreement was greater at 72.6% (95% CI, 61.9 to 85.7). When the LLM was stripped of information describing patients' values, its ability to choose the same treatment as the patient decreased substantially and matched the ability of human proxies. In secondary analyses, we explored the extraction of value profiles from electronic health record (EHR) data for patients who lose decisional capacity before formally documenting their values, and we critically evaluated value profiles written at lower reading levels. We found that the LLM inferred patients' values from real-world EHR clinical notes with low accuracy, underscoring the importance of having patients formally document their values before decisional capacity is lost. LLM utility degraded when value profiles were written at lower reading levels. Value-informed LLMs show promise as a proof of concept in the potential to assist human proxies in making treatment decisions that reflect the values of patients who have lost decisional capacity. Our findings caution against inferring patients' values from EHR data and using value profiles that are written at lower reading levels.

Victoria J. Nolan and Marcia MacGregor Brown contributed equally to this article.

Ray E. Moseley and Tyler J. Loftus contributed equally to this article.

The author affiliations are listed at the end of the article.

Tyler J. Loftus can be contacted at tyler.loftus@surgery.ufl.edu.

Introduction

When patients lose decision-making capacity, treatment decisions shift to a patient-designated, next-of-kin, or assigned proxy decision-maker. Ideally, proxies rely on prerecorded documentation like an advance directive (AD) to honor the patients' treatment preferences.[1,2] However, AD effectiveness has been questioned for several reasons:[2,3] almost two thirds of adults never complete one,[4] AD completion rates are lower for individuals with low health literacy or educational attainment,[5,6] and half of adults in the United States read below ninth-grade level, which may affect communication of treatment preferences.[7-9] Proxy decision-making is further complicated by the heavy cognitive and emotional burdens of the role,[10,11] inconsistent proxy engagement coordinating care,[12] and overconfidence in understanding patients' wishes.[13] Thus, well-intentioned proxy decisions are discordant with patients' wishes approximately half the time, often favor aggressive treatment that patients never desired, occasionally defer treatment that patients would have desired, and frequently impart moral distress on loved ones and clinicians.

Empirical investigations of large language models (LLMs) augmenting proxy decision-making are scarce and fraught with complex ethical dilemmas surrounding end-of-life care and decisional incapacity.[14,15] Our proof-of-concept simulation studies[16] suggest that LLMs may provide treatment recommendations that align with patients' values, but comparisons between LLM recommendations and choices by real human proxies are lacking. In addition, there is controversy regarding the prospect of extracting patients' values from electronic health record (EHR) data for situations in which patients lose decisional capacity before their values are formally documented, and, while literacy and education levels have well-documented associations with AD processes,[5,6,17] it remains unknown whether LLM-enabled decision support is adversely affected when text-based inputs are written at lower reading levels.

Our primary objective is to determine whether a value-informed, LLM-enabled proxy decision support framework can match or improve agreement with patient choices in hypothetical clinical scenarios. Our secondary objectives are to assess whether LLMs may be able to extract incapacitated patients' treatment preferences from EHR-based clinical notes (for situations in which patients lose capacity before their values are formally documented) and whether LLMs can maintain stable performance when prompted at lower reading levels. The resulting framework must augment rather than replace human proxies and must respect the deeply human and ethically sensitive nature of proxy decisions.

Methods

FRAMEWORK OVERVIEW AND LARGE LANGUAGE MODEL

Our full framework uses pairs of patients and the real-life legal health care proxy that the patient chose to make decisions for them in the event that the patient loses decisional capacity (patient–proxy dyads), patient value profiles (derived from standardized questionnaires completed by each patient), and LLM prompt engineering to generate value-concordant treatment recommendations for decisionally incapacitated patients. Our LLM of choice was the 8B-parameter Llama 3.1,[18] trained on over 15 trillion multilingual tokens of publicly available and human-annotated data sources, accessed through the Ollama application programming interface[19] (see Supplementary Appendix). Fine-tuning the temperature hyperparameter (as described in the Supplementary Appendix and illustrated in Figs. S1A and S1B) demonstrated that a value of 0.2 (higher precision, lower creativity)[20] generates the most performant outputs and, thus, this value was used in all prompts. We complied with Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines (Supplementary Appendix) and obtained institutional review board approval (IRB202400298, protocol # ET00023920).

DYAD EXPERIMENTS

We prospectively enrolled 30 community-dwelling adults (18 years of age or older) with decisional capacity, comprising 15 patient–proxy dyads (hereafter "dyad cohort"), each consisting of one subject in the role of "patient" alongside their actual legal proxy (illustrated in Fig. 1A and described in the Supplementary Appendix). Participants were recruited through academic partner referrals and snowball sampling.[21,22] Each patient completed the Value-Living Questionnaire (VLQ) and the Compassion and Choices (CC) worksheet (Supplementary Appendix). We chose these resources because they are standardized, publicly available, capture what matters to a person across life domains, and are less brittle than first-order treatment preferences (e.g., "I don't want to receive hemodialysis"), which fail to generalize to unanticipated clinical scenarios. These resources are also concrete enough to generate

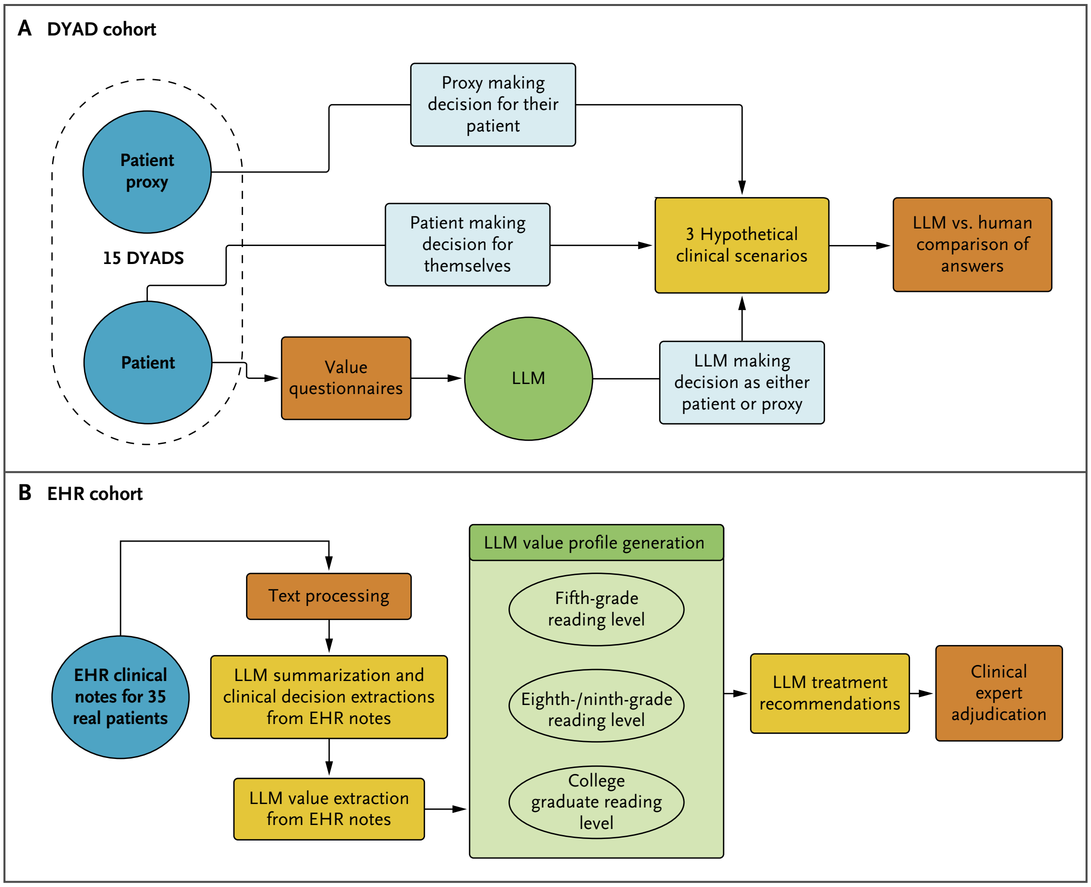

Figure 1. Data Processing and Modeling Framework for Assessing Value-Informed, Large Language Model–Enabled Decision Support.

Panel A compares large language model (LLM) and human proxy decision-making for hypothetical end-of-life scenarios with or without added knowledge of the patient's documented values. We asked the LLM to output one of five multiple-choice answers in response to three hypothetical decisional incapacity scenarios, answering as either the patient or the proxy. Panel B shows an outline of how we used text processing and prompt engineering to summarize and extract clinical information, create unstructured value profiles written at different reading levels, and obtain LLM treatment recommendations for decisionally incapacitated real-life patients. We manually adjudicated treatment responses, as well as LLM summarization and extraction accuracy. EHR denotes electronic health record; and LLM, large language model

actionable recommendations, therefore representing the closest approximation to current clinical standards of care in ascertaining values that could inform decisions regarding medical treatment. Our use of these two questionnaires to compose patient value profiles is, to our knowledge, a novel application.

We randomly selected and adapted eight simulated scenarios from our proof-of-concept study[16] and randomly allocated three to each dyad (Supplementary Appendix). Each scenario comprises one paragraph describing a hypothetical patient's background, clinical history, presenting complaints, reason for decisional incapacity, and treatment options. Scenarios were constructed to reflect the reality that patients' lived experiences are often dissimilar to those of the proxy, sometimes to an extreme, which may generate tension that compromises proxies' ability to offer substituted judgment by reporting to clinicians what a patient would have wanted. We asked each patient and proxy to choose one of five discrete treatments (Supplementary Appendix) for each scenario, blinded to their counterpart's answers: patients were to answer as though they were the patient in the scenario and select the treatment plan they would prefer for themselves, whereas proxies were to select the treatment plan that they thought best matched the patient's preferences. This arrangement was intended to reflect the clinical reality that when a patient loses decision-making capacity, clinicians ask the proxy to report to the clinicians what the patient would have wanted if the patient could have made the decision for themselves. Under that real-world circumstance, the ground truth — what the patient would choose for themselves — is unknown. Our experimental design uses hypothetical scenarios because this allows us to ascertain ground truth while the patient has decision-making capacity.

Proxies had access to any information about patients' preferences (from documents or conversations) provided to them by the patient who selected them as their legal proxy — as would occur in real-world circumstances — and we did not provide patients' value profiles or any other documents (including ADs) to the proxies, consistent with guidance from the American Bar Association[23] and the National Institute on Aging[24] that the responsibility for distributing these documents lies with the individual (patient). We then instructed the LLM to act as either patient or proxy and choose one of the same five treatments for the same three scenarios (Supplementary Appendix, Prompts 1–4), recognizing that LLM responses can vary substantially according to the assigned human role. Two prompts did not incorporate patient value information; two included a value profile created by concatenating 20 questions from the VLQ and CC with numeric answers (e.g., "How important do you consider family to be? Answer=10"). Full prompts are available in the Supplementary Appendix.

DATA ANALYSIS

Decisional agreement between patient and proxy or patient and LLM was calculated as the percentage of answers in which both parties selected the same treatment choice from among five discrete treatment options. Interim analysis demonstrated that LLMs never selected the fifth option, "Proxy to make this decision on his or her own," likely because we asked for a treatment recommendation. Thus, to avoid penalizing human proxies, the final analysis removed three dyads where this option was selected by the patient or proxy. Secondary analyses that explore the extraction of value profiles from EHR data for situations in which patients lose capacity before their values are formally documented are illustrated in Figure 1B and described in the Supplementary Appendix (cohort assembly and LLM-generated treatment recommendations). One board-certified clinical ethicist, two board-certified physicians, and three medical students adjudicated LLM outputs for the EHR cohort using Likert scoring[25] for agreement that outputs matched patient values and were clinically sensible, as previously described[16] (Supplementary Appendix). Adjudicators also rated LLM summarization and value profile creation processes on a 10-point accuracy scale (Supplementary Appendix). We assessed interrater reliability using Fleiss's kappa[26] (described in the Supplementary Appendix and illustrated in Table S1). We performed significance testing (Supplementary Appendix) using ANOVA (analysis of variance) and linear mixed-effects models. We calculated 95% confidence intervals (CIs) on all means using clustered bootstrapping (Supplementary Appendix).

Results

PRIMARY ANALYSIS: HUMAN PROXY VERSUS LARGE LANGUAGE MODEL AGREEMENT WITH PATIENT CHOICES

Average agreement between patients and their legal human proxies (i.e., choosing the same treatment in a hypothetical clinical scenario) was 44.7% (95% CI, 28.6 to 61.9) (Fig. 2). One third of the dyads selected different treatment plans in three of three scenarios: 17% of dyads chose matching responses for all three scenarios. Average agreement between patient and LLM was similar to patient–proxy agreement when no additional value information was provided to the LLM — 42.0% (95% CI, 32.4% to 51.3%) when answering as the patient and 45.2% (95% CI, 34.3% to 55.7%) when answering as the proxy (Fig. 2) — in the context that patients' value profiles were not provided to the human proxy by our investigator team. When patients' value profiles enriched the LLM, patient–LLM agreement increased to 72.6% (95% CI, 61.9% to 85.7%) and 72.9% (95% CI, 62.4% to 85.7%) when answering as the patient or proxy, respectively.

SECONDARY ANALYSIS: LARGE LANGUAGE MODEL PERFORMANCE AT EXTRACTING AND SUMMARIZING VALUES FROM ELECTRONIC HEALTH RECORD DATA

LLM accuracy in summarizing each patient's 10 most recent clinical notes had a mean adjudicator score of 7.7 of 10 (95% CI, 7.1 to 8.6). LLM accuracy in generating value profiles using information extracted from EHR notes had a mean adjudicator score of 7.63 of 10 (95% CI, 7.25 to 7.97). Adjudicator scoring demonstrated 74% agreement that LLM recommendations based on EHR records honored patients' value profiles (mean Likert score: 3.70/5, 3.55–3.85), and 79% agreement that LLM recommendations were clinically sensible[16] (mean Likert score: 3.93/5, 3.77–4.08).

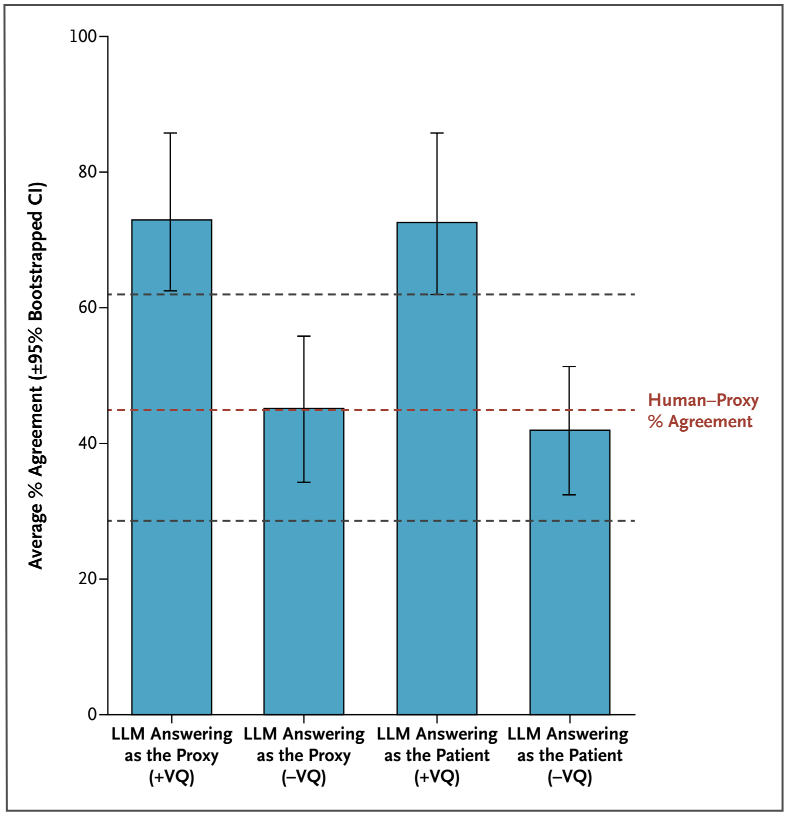

Figure 2. Average Large Language Model–Patient Agreement (Blue Bars) Compared with Average Human Proxy–Patient Agreement (Horizontal Brown and Black Dotted Lines) is Shown across 10 Repetition Trials (±95% Confidence Intervals).

Patients, proxies, and the large language model (LLM) were asked to select one of five multiple-choice answers to three hypothetical clinical scenarios, imagining it was either themselves (patient or LLM) or their patient (proxy or LLM) in the scenario. Average agreement was calculated as the percentage of times the proxy or LLM produced the same answer as the patient across the three scenarios and ranged from 0 (no matching answers) to 100 (all answers matched). LLM model prompts that include additional information from the patients' value questionnaires (VQs) (+VQ) showed higher agreement with patient responses than with LLM responses without (–VQ), regardless of answering as either the patient or proxy, and also showed higher agreement than with the human proxies. CI denotes confidence interval; LLM, large language model; and VQ, value questionnaire.

SECONDARY ANALYSIS: LARGE LANGUAGE MODEL PERFORMANCE ACROSS READING LEVELS

Human adjudications demonstrated that value profiles written at a fifth-grade reading level had lower accuracy than the eighth- or ninth-grade or college-level profiles (P<0.001) (see Fig. 3 and the Supplementary Appendix) and had lower performance in honoring patients' values and preferences (P<0.001) (see Fig. 4 and the Supplementary Appendix); however, interrater agreement was poor (see Figs. S2 and S3). These differences were observed despite sample consistency being high and similar across reading levels (see Fig. S4). Additional experiments (described in the Supplementary Appendix and illustrated in Fig. S5) found that the phenomenon of lower model performance when value profiles were written at lower reading levels was not seen when using simulated data: summarization accuracy and ability of the model to generate recommendations that honor patients' values were similar across reading levels when using simulated data.

Discussion

We study the potential utility of LLMs as decision support for decisionally incapacitated patients and their proxies using real patient–proxy dyads or EHR-derived clinical notes. We corroborate prior evidence that proxies demonstrate surprisingly low agreement with patients' wishes and preferences,

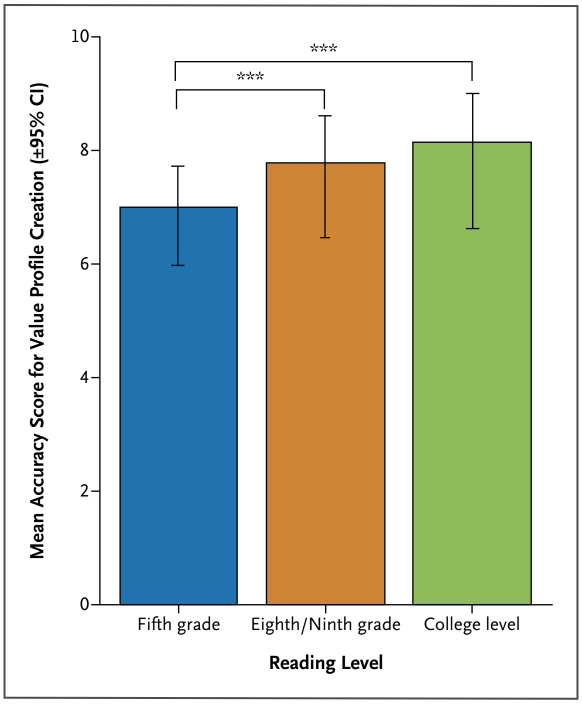

Figure 3. Mean Adjudicator Score (±95% Bootstrapped Confidence Intervals) Assessing the Accuracy of Large Language Model Value Profile Generation (Score Out of 10) for Each Patient in the Electronic Health Record Cohort across the Three Reading Levels: Fifth Grade, Eighth or Ninth Grade, and College Graduate Level.

The value profiles written at a fifth-grade reading level had lower accuracy than with eighth-/ninth-grade or college-level profiles (both P<0.001). *=P<0.05, **=P<0.01, and ***=P<0.001. CI denotes confidence interval.

even when the proxy is patient-designated rather than a next-of-kin or appointed proxy (e.g., hospital social worker), who may be less well equipped to understand the patient's values.[27-30] We show that LLMs, when provided with a simple summary of a patient's values, may have greater agreement with patients' choices than legal patient-designated proxies. Collectively, these findings suggest that LLM frameworks warrant future study as decision support tools to aid proxies and clinicians in the complex, emotionally burdensome task of decision-making for incapacitated patients.

When patients lack documented value profiles or ADs, clinical decision-making is arduous for both human proxies[30,31] and LLMs. When value profiles were omitted from our LLM framework, its performance was no better than that of human proxies. Thus, capturing accurate and informative value profiles is crucial to the effectiveness of LLM-enabled proxy decision support. This observation is consistent with other perspectives and evidence that human values are inherently represented in LLMs in ways that are highly consequential and poorly understood, thus requiring new methods to capture and codify human values.[32,33] We also show that LLMs may have low accuracy in extracting and summarizing patient values from noisy EHR clinical notes. Inferred values could theoretically be preferable to having no information about patients' values, as occurs in clinical settings when ADs and patient-designated and next-of-kin proxies are unavailable or ambiguous. However, inferring values with low accuracy could, through automation bias and undue authority that algorithmic outputs can exert on clinical decision-making, violate the bioethical principle of nonmaleficence and the Hippocratic teaching to "first, do no harm." The low accuracy with which patients' values were inferred from EHR data underscores the importance of having patients formally document their values before decisional capacity is lost.

High levels of health literacy and education are associated with increased AD completion rates,[17,34] a higher likelihood of discussing end-of-life decisions, and a higher likelihood of appointing a proxy.[35] Participants with low levels of health literacy and educational attainment show stronger preferences for aggressive care,[36] and consequently incur greater health care costs.[37] In addition, many AD forms are written at a 12th-grade reading level with complex terminology.[38] When ADs were tailored for reading level, their completion rates increased and the new format was preferred by 73% of participants.[5] We show that disparities among reading levels persist when applying our LLM framework.

STUDY LIMITATIONS

Our small sample sizes increase the risk for false-negative findings. Our findings are best seen as a proof of concept; a larger sample of participants with population-based probability sampling is needed. We observed discrepancies in the LLM choosing between non–mutually exclusive treatment options such as "Keep me comfortable and pain free" and "Do not perform cardiopulmonary resuscitation (CPR)." Thus, future work should ensure that treatment options are mutually exclusive so that proxy decision-support prompts are not confounded by multiple treatment options that may equally represent value-concordant care. In secondary analyses, Fleiss's kappa agreement tests showed no significant agreement among adjudicators, underscoring large variation among humans in complex decision-making processes. Conversely, LLM sample consistency[39] experiments showed LLM outputs were highly similar across model runs and output semantics were consistent. End-of-life care

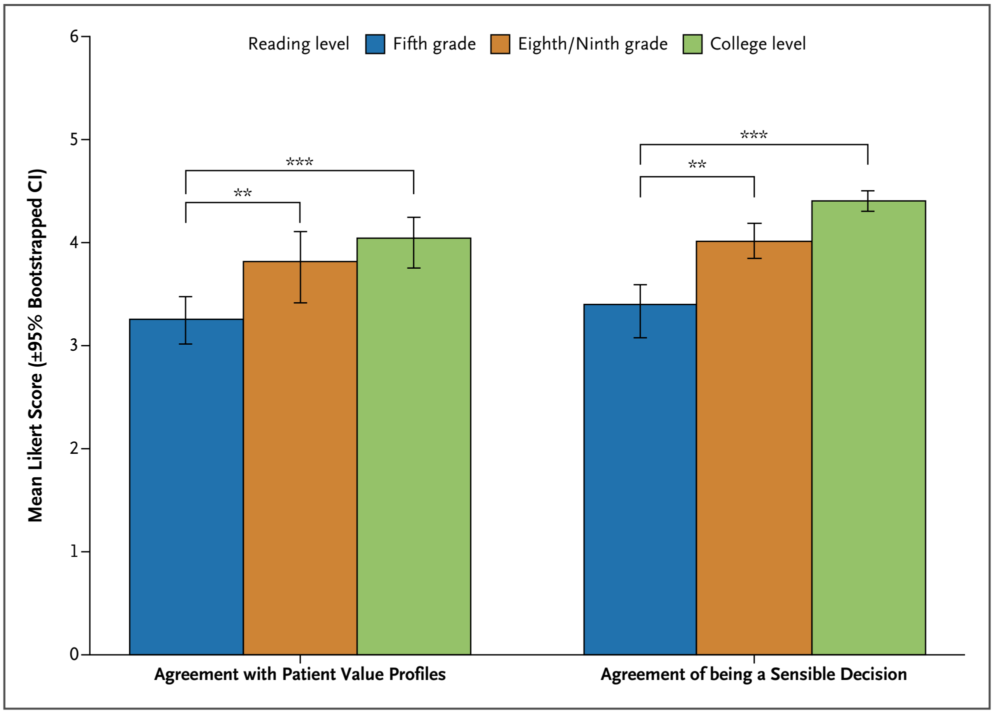

Figure 4. Adjudication of Large Language Model Treatment Recommendations at the Three Reading Levels (Fifth Grade, Eighth/Ninth Grade, or College Graduate Level) Averaged across Adjudicators and the 35 Electronic Health Record Cohort Patients.

Large language model (LLM) treatment recommendations were assessed using Likert scoring (0=no agreement, 5=perfect agreement) in accordance with: (1) did the recommended treatment match the values of the patient stated in the value profiles, and (2) was the recommended treatment a sensible and sound decision for the patient. Manual adjudication revealed that value profiles written at a fifth-grade reading level produced LLM treatment recommendations that performed worse at honoring patients' values or being sensible treatment decisions than those produced for higher reading levels (P<0.01). *=P<0.05, **=P<0.01, and ***=P<0.001. CI denotes confidence interval.

decisions often present interperson variability challenges, especially when patient values evolve.[40] LLM-enabled decision support implementation would require mechanisms for patients to dynamically update their value profiles (e.g., through electronic patient portals and during in-person health care encounters) and for LLMs to offer explanations for their outputs, which may alter the outputs themselves. Finally, the ethically and legally sensitive nature of LLM use in end-of-life decision-making requires transparency, accountability, and ongoing guidance from multidisciplinary human-in-the-loop stakeholders.

Conclusion

Value-informed LLM frameworks may be useful in assisting proxies in making treatment decisions that honor the values and preferences of decisionally incapacitated patients. Accurately and ethically capturing and understanding patients' values is paramount. Patients with decisional capacity may benefit by codifying their values so that when they lose capacity, loved ones and clinicians — aided by LLMs — may provide value-concordant care. We caution against the inference of patients' values from unstructured EHR data when documented expressions of values, ADs, and next-of-kin proxies are unavailable. Similar to previously observed associations between literacy levels and AD processes, we observe risks for using value profiles written at lower reading levels. LLM-enabled proxy decision support should augment rather than replace human proxies and must respect the deeply human and ethically sensitive nature of proxy decisions.

Disclosures

Author disclosures and other supplementary materials are available at ai.nejm.org.

All code used in performing this study is publicly available at: https://github.com/Prisma-pResearch/LLM_Proxy_Decision_Support.

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Author Affiliations

[1] Department of Surgery, University of Florida, Gainesville, FL, USA

[2] Department of Family, Youth, and Community Sciences, University of Florida, Gainesville, FL, USA

[3] College of Medicine, University of Florida, Gainesville, FL, USA

[4] Department of Medicine and Quantitative Health, College of Medicine, University of Florida, Gainesville, FL, USA

[5] Department of Philosophy, University of Hong Kong, Hong Kong

[6] Department of Surgery, University of Minnesota Medical School, University of Minnesota Twin Cities, Minneapolis, MN, USA

[7] Center for Learning Health Systems Science, University of Minnesota Medical School, University of Minnesota Twin Cities, Minneapolis, MN, USA

[8] Department of Community Health and Family Medicine, University of Florida, Gainesville, FL, USA

References

1. Emanuel LL, Barry MJ, Stoeckle JD, Ettelson LM, Emanuel EJ. Advance directives for medical care — a case for greater use. N Engl J Med 1991;324:889-895. DOI: 10.1056/NEJM199103283241305.
2. Sedini C, Biotto M, Crespi Bel'skij LM, Moroni Grandini RE, Cesari M. Advance care planning and advance directives: an overview of the main critical issues. Aging Clin Exp Res 2022;34:325-330. DOI: 10.1007/s40520-021-02001-y.
3. Perkins HS. Controlling death: the false promise of advance directives. Ann Intern Med 2007;147:51-57. DOI: 10.7326/0003-4819-147-1-200707030-00008.
4. Yadav KN, Gabler NB, Cooney E, et al. Approximately one in three US adults completes any type of advance directive for end-of-life care. Health Aff (Millwood) 2017;36:1244-1251. DOI: 10.1377/hlthaff.2017.0175.
5. Sudore RL, Landefeld CS, Barnes DE, et al. An advance directive redesigned to meet the literacy level of most adults: a randomized trial. Patient Educ Couns 2007;69:165-195. DOI: 10.1016/j.pec.2007.08.015.
6. Ladin K, Buttafarro K, Hahn E, Koch-Weser S, Weiner DE. "End-of-life care? I'm not going to worry about that yet." Health literacy gaps and end-of-life planning among elderly dialysis patients. Gerontologist 2018;58:290-299. DOI: 10.1093/geront/gnw267.
7. Kutner M, Greenberg E, Baer J. A first look at the literacy of America's adults in the 21st century. NCES 2006-470. Washington, DC: National Center for Education Statistics, 2005 (https://nces.ed.gov/use-work/resource-library/report/statistical-analysis-report/first-look-literacy-americas-adults-21st-century).
8. Luo Q, Shi K, Hung P, Wang S-Y. Associations between health literacy and end-of-life care intensity among Medicare beneficiaries. Am J Hosp Palliat Care 2021;38:626-633. DOI: 10.1177/1049909120988506.
9. Chiarchiaro J, Buddadhumaruk P, Arnold RM, White DB. Prior advance care planning is associated with less decisional conflict among surrogates for critically ill patients. Ann Am Thorac Soc 2015;12:1528-1533. DOI: 10.1513/AnnalsATS.201504-253OC.
10. Wendler D, Rid A. Systematic review: the effect on surrogates of making treatment decisions for others. Ann Intern Med 2011;154:336-346. DOI: 10.7326/0003-4819-154-5-201103010-00008.
11. Hickman RL Jr, Pinto MD. Advance directives lessen the decisional burden of surrogate decision-making for the chronically critically ill. J Clin Nurs 2014;23:756-765. DOI: 10.1111/jocn.12427.
12. Fried T, Zenoni M, Iannone L. A dyadic perspective on engagement in advance care planning. J Am Geriatr Soc 2017;65:172-178. DOI: 10.1111/jgs.14497.
13. Song MK, Ward SE, Lin FC. End-of-life decision-making confidence in surrogates of African-American dialysis patients is overly optimistic. J Palliat Med 2012;15:412-417. DOI: 10.1089/jpm.2011.0330.
14. Grote T, Berens P. On the ethics of algorithmic decision-making in healthcare. J Med Ethics 2020;46:205-211. DOI: 10.1136/medethics-2019-105586.
15. Biller-Andorno N, Ferrario A, Joebges S, et al. AI support for ethical decision-making around resuscitation: proceed with care. J Med Ethics 2022;48:175-183. DOI: 10.1136/medethics-2020-106786.
16. Nolan VJ, Balch JA, Baskaran NP, et al. Incorporating patient values in large language model recommendations for surrogate and proxy decisions. Crit Care Explor 2024;6:e1131. DOI: 10.1097/CCE.0000000000001131.
17. Chen DR, Weng HC. Associations of health literacy, personality traits, and pro-individualism with the willingness to complete advance directives in Taiwan. BMC Palliat Care 2023;22:91. DOI: 10.1186/s12904-023-01215-3.
18. Meta AI. Introducing Meta Llama 3: the most capable openly available LLM to date. April 18, 2024 (https://ai.meta.com/blog/meta-llama-3/).
19. Ollama. Download Ollama. 2024 (https://ollama.com/download).
20. Peeperkorn M, Kouwenhoven T, Brown D, Jordanous A. Is temperature the creativity parameter of large language models? In: Proceedings of the 15th International Conference on Computational Creativity. Jönköping, Sweden: ICCC, 2024:226-235. DOI: 10.48550/arXiv.2405.00492.
21. Kennedy-Shaffer L, Qiu X, Hanage WP. Snowball sampling study design for serosurveys early in disease outbreaks. Am J Epidemiol 2021;190:1918-1927. DOI: 10.1093/aje/kwab098.
22. Social Science Statistics. Effect Size Calculator (Cohen's d). (https://www.socscistatistics.com/effectsize/default3.aspx).

23. American Bar Association. Health care decision-making. November 8, 2023 (https://www.americanbar.org/groups/law_aging/resources/health_care_decision_making/).

24. National Institute on Aging. Advance care planning: advance directives for health care. October 31, 2022 (https://www.nia.nih.gov/health/advance-care-planning/advance-care-planning-advance-directives-health-care).

25. Likert R. A technique for the measurement of attitudes. Arch Psychol 1932;22:55.

26. Fleiss JL. Measuring nominal scale agreement among many raters. Psych Bull 1971;76:378-382. DOI: 10.1037/h0031619.

27. Shalowitz DI, Garrett-Mayer E, Wendler D. The accuracy of surrogate decision makers: a systematic review. Arch Intern Med 2006;166:493-497. DOI: 10.1001/archinte.166.5.493.

28. Bibas L, Peretz-Larochelle M, Adhikari NK, et al. Association of surrogate decision-making interventions for critically ill adults with patient, family, and resource use outcomes: a systematic review and meta-analysis. JAMA Network Open 2019;2:e197229. DOI: 10.1001/jamanetworkopen.2019.7229.

29. Comer AR, Hickman SE, Slaven JE, et al. Assessment of discordance between surrogate care goals and medical treatment provided to older adults with serious illness. JAMA Network Open 2020;3:e205179. DOI: 10.1001/jamanetworkopen.2020.5179.

30. Silveira MJ, Kim SY, Langa KM. Advance directives and outcomes of surrogate decision making before death. N Engl J Med 2010;362:1211-1218. DOI: 10.1056/NEJMsa0907901.

31. Bravo G, Sene M, Arcand M. Making medical decisions for an incompetent older adult when both a proxy and an advance directive are available: which is more likely to reflect the older adult's preferences? J Med Ethics 2018;44:498-503. DOI: 10.1136/medethics-2017-104203.

32. Yu K-H, Healey E, Leong T-Y, Kohane IS, Manrai AK. Medical artificial intelligence and human values. New Engl J Med 2024;390:1895-1904. DOI: 10.1056/NEJMra2214183.

33. Kohane IS, Manrai AK. The missing value of medical artificial intelligence. Nat Med 2025;31:3962-3963. DOI: 10.1038/s41591-025-04050-6.

34. Waite KR, Federman AD, McCarthy DM, et al. Literacy and race as risk factors for low rates of advance directives in older adults. J Am Geriatr Soc 2013;61:403-406. DOI: 10.1111/jgs.12134.

35. Meier C, Vilpert S, Wieczorek M, Borrat-Besson C, Jox RJ, Maurer J. End-of-life health literacy, knowledge and behaviours towards advance care planning among older adults: cross-sectional evidence from Switzerland. BMJ Public Health 2024;2:e000600. DOI: 10.1136/bmjph-2023-000600.

36. Volandes AE, Paasche-Orlow M, Gillick MR, et al. Health literacy not race predicts end-of-life care preferences. J Palliat Med 2008;11:754-762. DOI: 10.1089/jpm.2007.0224.

37. Shi K, Hung P, Wang SY. Associations among health literacy, end-of-life care expenditures, and rurality. J Rural Health 2021;37:517-525. DOI: 10.1111/jrh.12513.

38. Ott BB, Hardie TL. Readability of advance directive documents. Image J Nurs Sch 1997;29:53-57. DOI: 10.1111/j.1547-5069.1997.tb01140.x.

39. Savage T, Wang J, Gallo R, et al. Large language model uncertainty proxies: discrimination and calibration for medical diagnosis and treatment. J Am Med Inform Assoc 2025;32:139-149. DOI: 10.1093/jamia/ocae254.

40. Auriemma CL, Nguyen CA, Bronheim R, et al. Stability of end-of-life preferences: a systematic review of the evidence. JAMA Intern Med 2014;174:1085-1092. DOI: 10.1001/jamainternmed.2014.1183.